

Office Cafeteria Management System

Problem Statement

Many organizations provide cafeteria facilities for employees to order breakfast, lunch, snacks, and beverages. However, meal ordering is often managed through manual processes such as phone calls, spreadsheets, or paper-based systems, leading to inaccurate orders, long waiting times, and unnecessary food wastage.

Without a centralized cafeteria management system, organizations face several challenges:

- Employees experience delays while placing meal orders.
- Cafeteria staff struggle to estimate meal demand accurately.
- Food wastage increases due to over-preparation.
- Employees have limited visibility into available menu items.
- Manual billing and reporting consume additional administrative effort.
- Lack of historical data makes demand forecasting difficult.

To improve efficiency and reduce food waste, organizations require a centralized **Office Cafeteria Management System** that allows employees to browse menus, place meal orders, and enables cafeteria administrators to manage menus, monitor orders, and generate sales reports.

Challenge

Design and develop an **Office Cafeteria Management System** that enables employees to conveniently order meals through a web application while providing cafeteria administrators with tools to manage menus, monitor orders, and analyze daily sales.

The application should provide a secure, user-friendly, and efficient platform for meal ordering, order management, reporting, and cafeteria operations.

Objectives

Develop a secure and centralized web application that enables organizations to:

- Allow employees to browse daily cafeteria menus.
 - Enable employees to place meal orders online.
 - Maintain order history for employees.
 - Allow administrators to manage menu items.
 - Generate daily sales reports.
 - Reduce food wastage through better demand visibility.
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User Roles

Administrator / Cafeteria Manager

The Administrator is responsible for managing menu items, orders, and cafeteria operations.

Responsibilities

- Add, update, and remove menu items.
 - View all employee orders.
 - Update order status.
 - Generate daily sales reports.
 - Monitor cafeteria performance through dashboards.
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Employee

Employees can browse available meals and place food orders.

Responsibilities

- Secure Login.
 - View cafeteria menu.
 - Place meal orders.
 - View current order status.
 - Access previous order history.
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Functional Requirements

1. Authentication

- Secure Login for Employees and Administrators.
 - Role-based authentication using Spring Security and JWT.
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2. Employee Login

Employees should be able to:

- Login securely.
 - View available menu items.
 - Place meal orders.
 - Track order status.
 - Access order history.
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3. View Menu

Employees should be able to browse the daily cafeteria menu.

Menu Details

- Item Name
- Category (Breakfast, Lunch, Snacks, Beverages)
- Description
- Price
- Availability Status
- Image (*Optional*)

The menu should only display currently available items.

4. Place Orders

Employees should be able to:

- Select one or more menu items.
- Specify quantity.
- Review order summary.

- Confirm the order.

Order Information

- Order ID
- Employee Name
- Order Date
- Ordered Items
- Quantity
- Total Amount
- Payment Status (*Optional*)
- Order Status

Order Status Options

- Placed
 - Preparing
 - Ready for Pickup
 - Completed
 - Cancelled
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5. Order History

Employees should be able to:

- View previous orders.
- Check order details.
- Filter orders by date.
- View order status.

Order history should include:

- Order Date
 - Ordered Items
 - Total Amount
 - Payment Status
 - Order Status
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6. Admin Menu Management

Administrators should be able to:

- Add new menu items.
 - Update existing menu items.
 - Delete menu items.
 - Mark items as available or unavailable.
 - Update pricing.
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7. Daily Sales Report

The system should generate reports containing:

- Total Orders
- Total Revenue
- Most Ordered Items
- Average Order Value
- Orders by Category
- Cancelled Orders

Optional Dashboard Analytics

- Daily Sales Trend
 - Weekly Revenue
 - Popular Menu Items
 - Peak Ordering Hours
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8. Search & Filter

Search By

- Menu Item
- Employee Name
- Order ID

Filter Options

- Today's Orders
- Completed Orders
- Cancelled Orders
- Breakfast Orders

- Lunch Orders
 - Snack Orders
 - Beverage Orders
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Database Design

Employee

Field

Employee ID

Name

Email

Department

Role

Password

Menu Item

Field

Item ID

Item Name

Category

Description

Price

Availability

Order

Field

Order ID

Employee ID

Order Date

Total Amount

Payment
Status

Order Status

Order Item

Field

Order Item ID

Order ID

Menu Item ID

Quantity

Price

Technology Stack

Backend

- Java 17
- Spring Boot
- Spring Data JPA
- Spring Security
- JWT Authentication
- REST APIs

Frontend

- React
- React Router
- Axios
- Material UI / Bootstrap

Database

- MySQL
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Bonus Features (*Optional*)

- Mock online payment integration.
 - QR code for order pickup.
 - Email notifications for order confirmation and status updates.
 - Estimated preparation time display.
 - Dashboard analytics with charts.
 - Export daily sales reports to Excel or PDF.
 - Meal pre-order for future dates.
 - Favorite menu items and quick reorder functionality.
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Evaluation Criteria

Criteria	Marks
Problem Understanding	10
UI/UX Design	15
Backend API Design	20
Database Design	15
Authentication & Role Management	10
CRUD Operations	10
Order Workflow & Validation	10
Innovation / Bonus Features	10

Presentation & Demo	10
Total	100

Expected Outcome

By the end of the hackathon, participants should deliver a responsive and secure **Office Cafeteria Management System** that streamlines meal ordering, menu management, and cafeteria operations while reducing food wastage through better demand planning. The solution should demonstrate core **Java Full Stack Development** concepts, including RESTful API development, role-based authentication, order management workflows, relational database design, and a responsive React-based user interface, while addressing a common operational need in modern organizations.